Arista Access Node MAR-ACC-45 QoS Congestion Event (Microbursts, Interface Tail Drops, QoS Queue Starvation) Resolution SOP

Node: MAR-ACC-45 | **Vendor:** Arista | **Severity:** Minor | **Alarm Code:** ALARM_4089

Description

This Standard Operating Procedure is designed to address QoS congestion events, specifically microbursts, interface tail drops, and QoS queue starvation on the Arista Access Node MAR-ACC-45 located at Marathahalli Core. This failure domain occurs due to the node's role as an access point, where the sheer volume of incoming traffic can overwhelm the available bandwidth, leading to congestion events. In Arista's eOS, QoS is implemented through the use of traffic classes, which are mapped to specific policy maps. These policy maps define the QoS parameters, including bandwidth allocation, packet dropping, and queuing. However, in the event of a microburst or interface tail drop, the QoS system may become overwhelmed, leading to queue starvation and further congestion. This SOP aims to identify the root cause of the congestion event and implement corrective actions to mitigate the issue.

Pre-requisite Checks

- Pre-requisite: Verify that the Marathahalli Core site is accessible and that the remote hands are dispatched to perform the necessary actions.

- Pre-requisite: Check the status of the redundant link to ensure that it is operational and can be used as a fallback in case of a failure.

- Pre-requisite: Review the node's configuration to ensure that the QoS policy maps are properly configured and that there are no configuration errors.

Potential Root Causes

Cause	Probability
Misconfigured QoS policy maps	Medium
Insufficient bandwidth allocation	High
Hardware issues with the interface or switch	Low

Diagnostic Commands & Expected Outputs

```
Command: show qos policers, Expected Output: Policers: 1 policer1: 1000 bps, Description: This command displays the configured policers on the node. In this case, we are checking to see if the policer is properly configured and if there are any configuration errors.
```

```
Command: show interface counters, Expected Output: Interface: 1/1/1 packets in: 1000000 packets out: 500000, Description: This command displays the interface counters for the node. We are checking to see if there are any indications of congestion or high packet loss on the interface.

Command: show qos policy maps, Expected Output: Policy Maps: 1 policy1: 1000 bps, Description: This command displays the configured policy maps on the node. We are checking to see if the policy map is properly configured and if there are any configuration errors.
```

Resolution Steps

Step 1: Step: Verify the QoS policy maps, Command: show qos policy maps, Description: Verify that the policy maps are properly configured and that there are no configuration errors. If there are any errors, correct them and reapply the policy maps.

Step 2: Step: Adjust the bandwidth allocation, Command: config qos policers policer1 2000 bps, Description: Increase the bandwidth allocation to 2000 bps to alleviate congestion on the interface.

Step 3: Step: Verify the interface counters, Command: show interface counters, Description: Verify that the packet counts on the interface have decreased and that there are no indications of congestion.

Post-Check Verification

```
Command: show qos policers, Expected Output: Policers: 1 policer1: 2000 bps, Description: Verify that the policer is properly configured and that the bandwidth allocation has been increased to 2000 bps.

Command: show interface counters, Expected Output: Interface: 1/1/1 packets in: 500000 packets out: 250000, Description: Verify that the packet counts on the interface have decreased and that there are no indications of congestion.
```

Roll-back Procedures

- Step: Reapply the original policer configuration, Command: config qos policers policer1 1000 bps, Description: Reapply the original policer configuration to restore the previous settings.

- Step: Verify the QoS policy maps, Command: show qos policy maps, Description: Verify that the policy maps are properly configured and that there are no configuration errors.